MOHAMED CHERIF

Principal / Staff Software Engineer, GPU-Accelerated Media, Computer Vision, and AI Systems

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PROFESSIONAL SUMMARY

Principal-level engineer with 20+ years building GPU-accelerated media pipelines, SDKs, and computer vision systems at AMD, Microsoft, Amazon, and Skype. ExpertiseC/C++ performance optimization, GPU programming (CUDA/ROCm), plus modern ML engineering (Transformers, RAG, LLMs, PyTorch, Docker/CI). Proven record translating research prototypes into production systems.

PROFESSIONAL EXPERIENCE

AMD: Principal Member of Technical Staff - *London, UK | Feb 2022 – Jan 2025*

- Architected and led SmartAccess Video: AMD's distributed GPU processing SDK. Achieving >60% transcoding speed improvements with intelligent workload distribution across integrated/discrete GPUs
- Drove prototypes to production by collaborating with engineering teams and ISV partners to deliver measurable quality/performance gains in shipping products. Received Executive Spotlight Award.
- Owned performance profiling/optimization loops across CPU/GPU, including memory and scheduling bottlenecks, using GPUView/CodeXL/VTune and custom telemetry.

TeamViewer: Technical Manager, Business Development - *Stuttgart, Germany | May 2019 – May 2020*

- Grew enterprise integrations 25% through technical evangelism, MDM/EMM portal partnerships, and developer enablement programs
- Supported enterprise accounts in Digital Twins/IoT implementations, advising on REST API architecture for multi-tenancy and unattended access

Amazon Prime Video: Senior Technical Program Manager - *London, UK | Jul 2016 – Mar 2018*

- Led end-to-end integration of Amazon Video Player on living-room SoC devices, supporting Prime Video's launch in 200+ countries
- Delivered next-generation players with DASH/HEVC support and enhanced monetization through optimized ad insertion while maintaining playback quality

Skype: Senior Program Manager, Real-Time Media - *Stockholm, Sweden | Nov 2013 – Oct 2015*

- Owned feature development and optimization across Skype's real-time communication pipeline, coordinating cross-functional teams and hardware partners
- Drove performance improvements in video encoding/decoding through collaboration with IHV/OEM partners on hardware-accelerated implementations

AMD: Senior Member of Technical Staff & Team Lead - *Toronto, Canada | Nov 2003 – Jun 2012*

- Led team of 8-12 engineers delivering AMD's GPU-based video codec pipeline from pre-silicon emulation through production, supporting worldwide sales and ISV partnerships
- Architected and implemented H.264, MPEG-2, VC-1, MVC, H.263, and MPEG-4 codec components; achieved >3dB rate control improvements through encoder optimization
- Developed advanced error recovery and concealment algorithms for video decoding firmware across multiple ASIC generations

Independent / Consulting: Principal Software Engineer *(Selected engagements) | Jul 2012 – Present*

Selected client engagements across media, AR/VR, and applied CV/ML (overlapping with full-time roles):

- Delivered low-latency AR/VR streaming infrastructure for Volta's interactive mixed-reality platform

- Optimized Microsoft Xbox video pipeline to meet real-time gaming latency targets; built GPU-accelerated image/video filtering and visual effects
- Redesigned video scene analysis pipeline: reduced faulty detections from ~50% to ~5% and improved processing speed 10x
- Deployed automated content moderation across Schibsted Media marketplaces, integrating ML models for ad safety and private messaging

CORE SKILLS

- Low-level GPU & Systems: C/C++, CUDA, ROCm/HIP, memory hierarchy, kernel/runtime optimization; device drivers, firmware, HALs
- Perf & Debugging: GPUView, AMD CodeXL, Intel VTune, hardware emulators, custom instrumentation
- Media Pipelines: real-time encode/decode/transcode/filtering; codecs (H.26x, HEVC, MPEG-x, VC-1, MVC); pre-silicon to production
- Graphics & Platforms: DirectX, Vulkan; SoC bring-up; hardware partner enablement (IHV/OEM/ISV)
- CV & ML Systems: RTMDet, SAM, RVM; LLMs/RAG; PyTorch deployment; benchmarking/optimization
- Leadership: SDK/API design, technical roadmap, cross-functional delivery, partner integrations

SELECTED AI/ML PROJECTS & CERTIFICATIONS

AI Job Search Agent: Career Copilot **(2026)**

Built an autonomous job discovery & application agent that ingests listings from 20+ sources, de-duplicates across feeds, and evaluates fit using hybrid rule-based filters and local LLM reasoning (Ollama).
Added a tool-calling assistant, FastAPI UI, scheduler, and Playwright automation to support ATS applications and job workflow management.

Plant Disease RAG Assistant (2025)

Retrieval-Augmented Generation system for plant disease diagnosis and treatment guidance (vector embedding, and LLM orchestration). Production web demo and mobile app also on Hugging Face Spaces.

AI Travel Agent (2026)

Anam's persona-driven conversational AI with tool integration and live 3D terrain visualization.
Demonstrates modern LLM orchestration, function calling, and interactive UI choreography.

EdgeScope Studio (2025)

Local-first computer vision lab for image detection/segmentation (RTMDet+SAM) and real-time video matting/background blur (RVM) with reproducible benchmarks and CI.

Fine-Tuning & Reinforcement Learning for LLMs, Agentic AI, AI Agents in LangGraph, DeepLearning.AI (2026)

Implemented LoRA fine-tuning pipelines, RLHF rollouts, and evaluation workflows for LLM alignment tasks.
Agentic AI design patterns (reflection, planning, tool use, multi-agent collaboration).
Multi-agent LLM workflows using LangGraph and tool-based reasoning loops.

LLM ZoomCamp, MLOps ZoomCamp, DataTalks.Club (2025)

Built production-style LLM systems with RAG pipelines, vector search, and model evaluation.
Implemented ML pipelines with MLflow, Docker, CI/CD, experiment tracking, and monitoring.

EDUCATION

- M.A.Sc., Electrical & Computer Engineering: University of British Columbia, Canada (1997–2000)
- B.Eng. (Diplôme d'Ingénieur), Signals & Systems: Tunisia Polytechnic School (1994–1997)

PATENTS

- US 20130163670 A1: Multiview Video Coding Reference Picture Selection
- US 20130050414 A1: Navigating and Selecting Objects within a 3D Video Image
- US 20130301725 A1: Efficient Mode Decision for Multiview Video Coding